

Exercise 5: Restaurant Scenario

A restaurant wants to replace their paper-based system for tracking reservations, customers, and employees with a digital database.

Here is a bullet list of information that a user has provided about the restaurant and how it operates.

- The restaurant has exactly 20 physical dining tables
- Each table seats up to 6 customers
- Each table has a status: either "priority" or "non-priority"
- Customers can make reservations for any day of the year
- One customer can reserve multiple tables (up to 20) for the same day
- The restaurant has only one seating time per evening, so you only need to track the date (not specific times)
- All employees are wait staff who serve tables
- Employees have experience levels: Novice, Experienced, or Very Experienced
- Table assignment limits based on experience:
 - Novice: maximum 3 tables per day
 - Experienced: maximum 5 tables per day
 - Very Experienced: maximum 7 tables per day
- Only Very Experienced employees can work priority tables
- Table assignments can change daily for each employee
- Every table must be assigned to an employee each day
- At least one Very Experienced employee must work each day

Create a complete ER/EER model for the scenario above. You must use the symbols presented in class for your ER/EER model. Use inheritance if appropriate. Feel free to use abbreviations for entity, attribute, and relationship names. Remember, the scenario description may include information that is good to know but not necessarily data that needs to be stored. You are modeling data not how someone might want to use the data.

Some of the information may be in a Business Constraints, not necessarily in the graphical part of the ERD.